CM 2083
Regression Analysis

GROUP NO: 01

CONCRETE COMPRESSIVE STRENGTH

Analysis of Concrete Compressive Strength Using Regression Analysis

Acknowledgment

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1. Introduction

- Concrete is a composite material made from cement, water, and aggregates like sand or gravel. When mixed, the cement reacts with water to form a paste that binds the aggregates, hardening over time through curing. This gives concrete its strength and durability, making it one of the most widely used materials in construction. Its versatility allows it to be used in structures like buildings, roads, and bridges. Accurately predicting concrete's compressive strength is crucial for ensuring the safety and longevity of these structures.

1.1 Problem Statement

- Develop a predictive multiple linear regression model to accurately estimate the compressive strength of concrete based on key input features, such as cement, water, coarse aggregate, fine aggregate, and age.

- The model aims to ensure reliable predictions, supporting safer and more durable construction practices.

2. Data Description

2.1 About Dataset

- Data source: https://www.kaggle.com/datasets/niteshyadav3103/concrete-compressive-strength

- Data Characteristics:
  - Number of instances (observations): 1030
  - Number of Attributes: 9
  - Attribute breakdown: 8 quantitative input variables, and 1 quantitative output variable

2.2 Overview of the Dataset

Variable Name	Data Type	Measurement	Description
Cement	Quantitative	kg/m^3	The amount of cement in the concrete mixture.
Blast Furnace Slag	Quantitative	kg/m^3	The amount of blast furnace slag in the concrete mixture.
Fly Ash	Quantitative	kg/m^3	The amount of fly ash in the concrete mixture.
Water	Quantitative	kg/m^3	The amount of fly ash in the concrete mixture.
Superplasticizer	Quantitative	kg/m^3	The amount of superplasticizer in the concrete mixture.
Coarse Aggregate	Quantitative	kg/m^3	The amount of coarse aggregate in the concrete mixture.
Age (days)	Quantitative	days (1-365)	The curing age of the concrete.
Fine Aggregate	Quantitative	Kg/m^3	The amount of fine aggregate in the concrete mixture.

Compressive Strength	Quantitative	MPa	The target variable represents the concrete's compressive strength.

- Dependent Variable: Concrete Compressive Strength (y)

Independent Variables:

- Cement (x_1)
- Blast Furnace Slag (x_2)
- Fly Ash (x_3)
- Water (x_4)
- Superplasticizer (x_5)
- Coarse Aggregate (x_6)
- Age (days) (x_7)
- Fine Aggregate (x_8)

3. Data Cleaning

- Data cleaning is the process of detecting, correcting, or removing inaccuracies and inconsistencies from raw data to improve its quality and ensure it is ready for analysis. It is a crucial step in data preparation that ensures the dataset is accurate, complete, consistent, and free from errors that could affect the results of any analysis or machine learning models.

Key Aspects Data cleaning:

3.1 Handling Missing Data

- Handling missing values in datasets is crucial for ensuring data quality and preventing biases or inaccuracies in analysis, as it addresses gaps or null values due to human error, system malfunction, or incomplete data collection.

The **code** provided checks for missing values in both rows and columns.

```
#check the missing values
colSums(is.na(cement))
rowSums(is.na(cement))
```

Output:

```
> colSums(is.na(cement))
                      Cement            Blast Furnace Slag
                           0                             0
                     Fly Ash                         Water
                           0                             0
             Superplasticizer             Coarse Aggregate
                           0                             0
               Fine Aggregate                     Age (day)
                           0                             0
Concrete compressive strength
                           0
```

3.2 Removing Duplicates

- After dealing with missing values, the next step involves identifying and removing duplicate rows to ensure data consistency. According to Dataset, there are 25 duplicate rows.

Code

```
#1.3remove duplicates
#view the duplicated rows
duplicates <- duplicated(cement)

# View duplicate rows
cement[duplicates, ]

cement_clean=cement[!duplicated(cement),]
cement_clean

View(cement_clean)
```

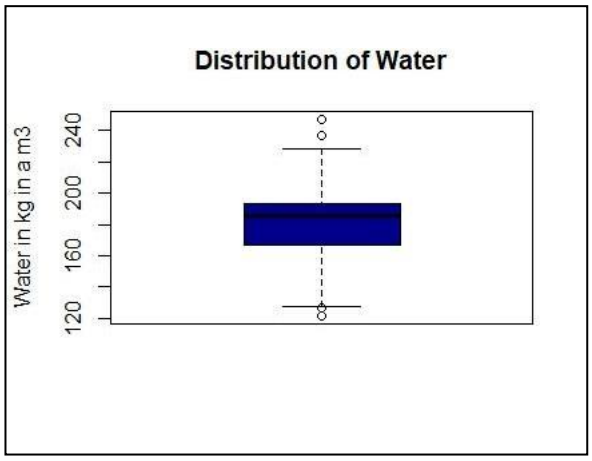
Output

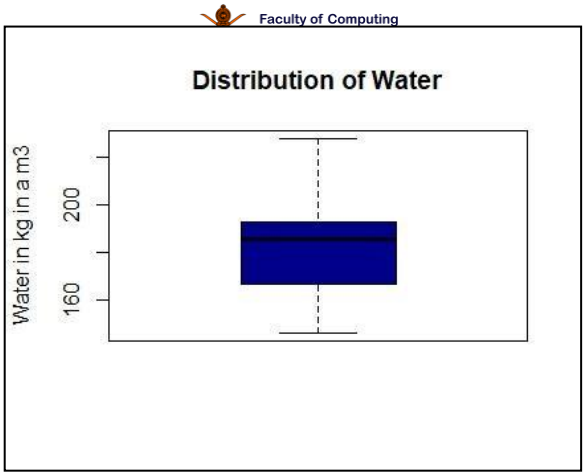
```
> #1.3remove duplicates
> #view the duplicated rows
> duplicates <- duplicated(cement)
> cement[duplicates, ]
# A tibble: 25 × 9
   Cement `Blast Furnace Slag` `Fly Ash` Water Superplasticizer `Coarse Aggregate`
    <dbl>                <dbl>     <dbl> <dbl>            <dbl>              <dbl>
 1   425                  106.         0  154.             16.5               852.
 2   425                  106.         0  154.             16.5               852.
 3   363.                 189          0  165.             11.6               945.
 4   363.                 189          0  165.             11.6               945.
 5   363.                 189          0  165.             11.6               945.
 6   425                  106.         0  154.             16.5               852.
 7   425                  106.         0  154.             16.5               852.
 8   363.                 189          0  165.             11.6               945.
 9   363.                 189          0  165.             11.6               945.
10   425                  106.         0  154.             16.5               852.
# ℹ 15 more rows
# ℹ 3 more variables: `Fine Aggregate` <dbl>, `Age (day)` <dbl>,
#   `Concrete compressive strength` <dbl>
# ℹ Use `print(n = ...)` to see more rows
```

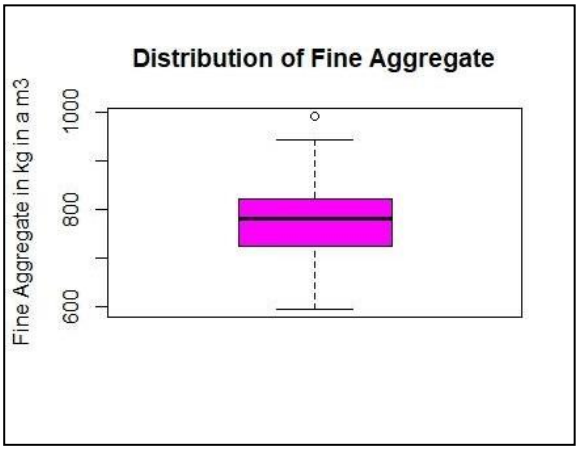
3.3 Dealing with Outliers

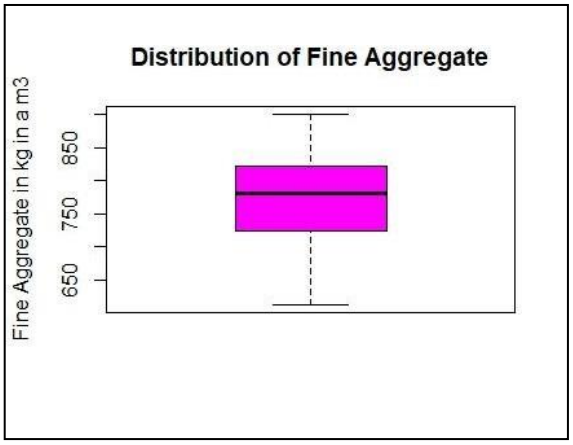
- Outliers are extreme values that significantly differ from the rest of the data, often due to measurement errors, variability, or rare events, and can impact statistical analyses or machine learning models. Here we used several methods of handling outliers.

1. Winsorization - Winsorization is a statistical technique used to limit the influence of extreme values (outliers) in a dataset by replacing them with less extreme values.

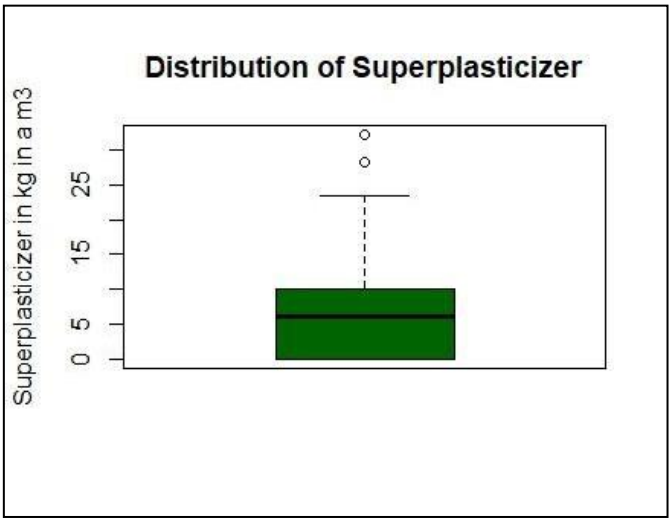


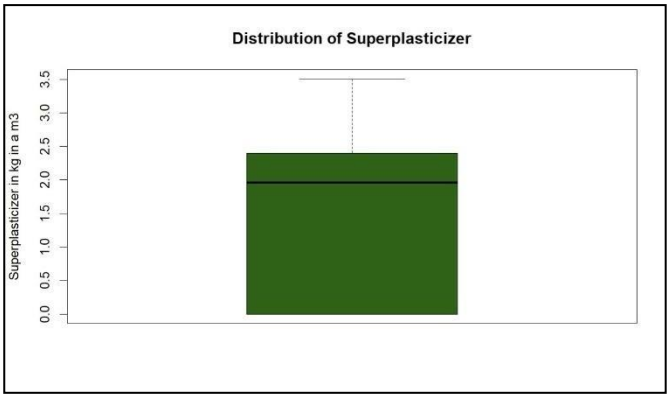


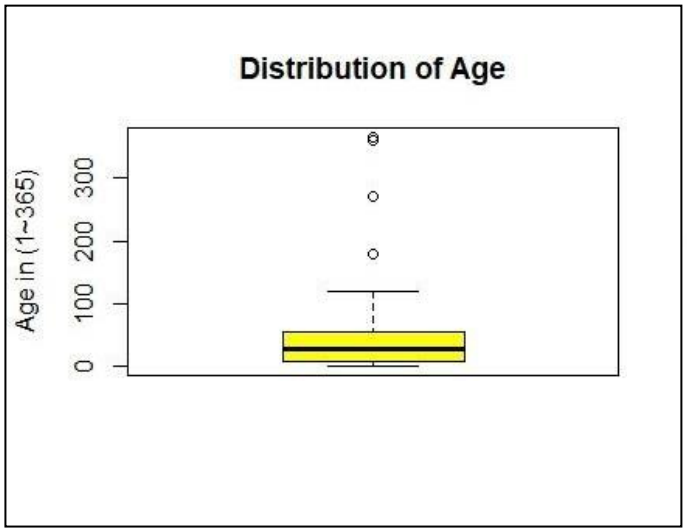


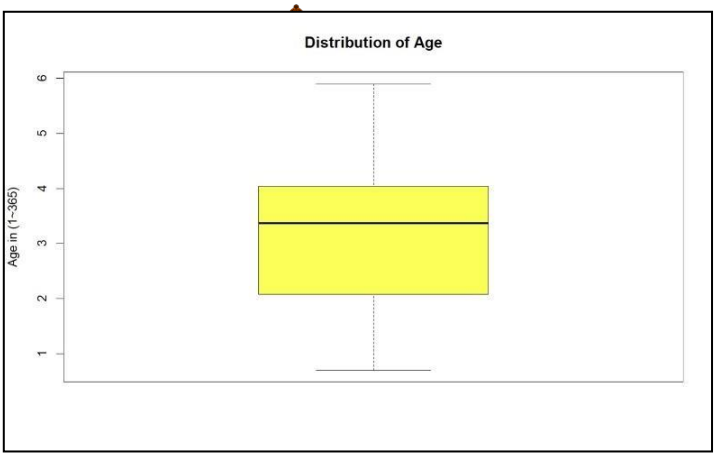


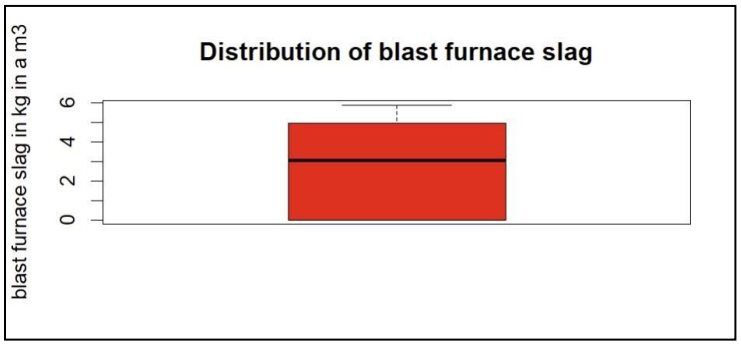
2. Log Transformation - By applying a log transformation, the extreme values (outliers) are compressed and brought closer to the rest of the data, reducing their impact on the analysis.

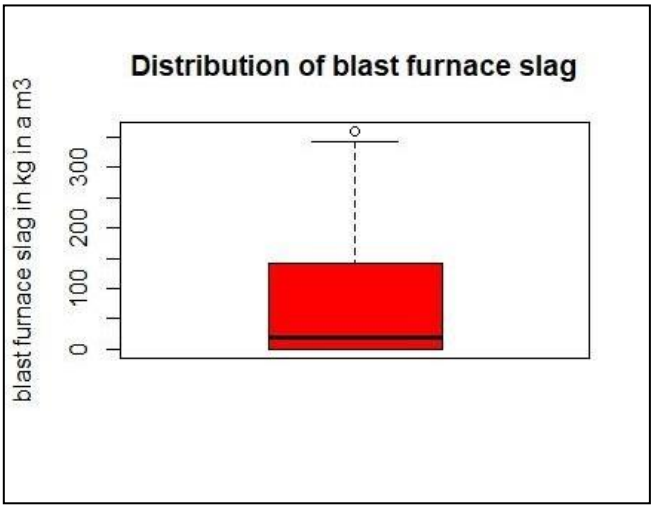




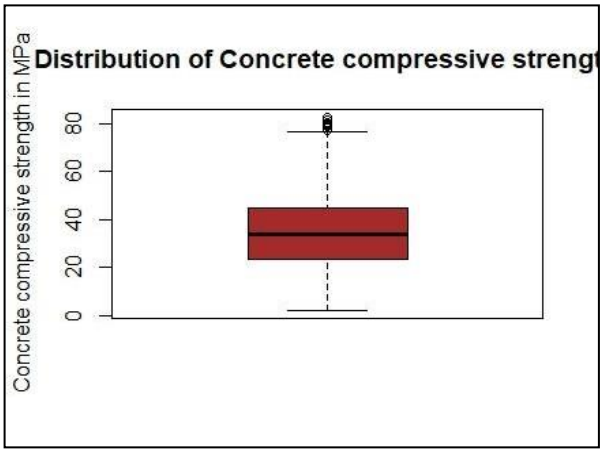


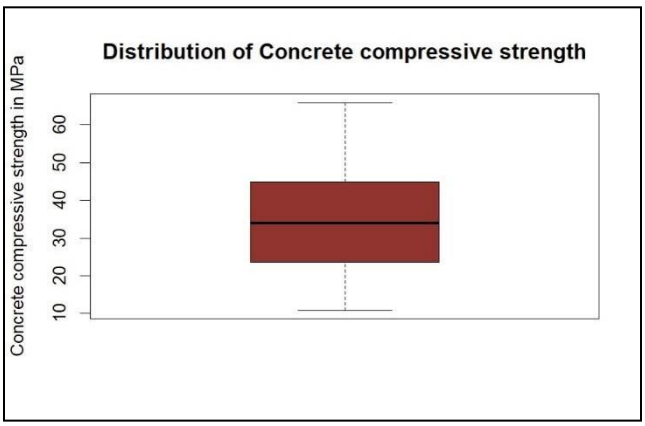






3. Capping - Setting upper and lower limits (or caps) for the data, beyond which extreme values (outliers) are "capped" to these limits. Instead of removing outliers, their values are adjusted to be within a specified range.



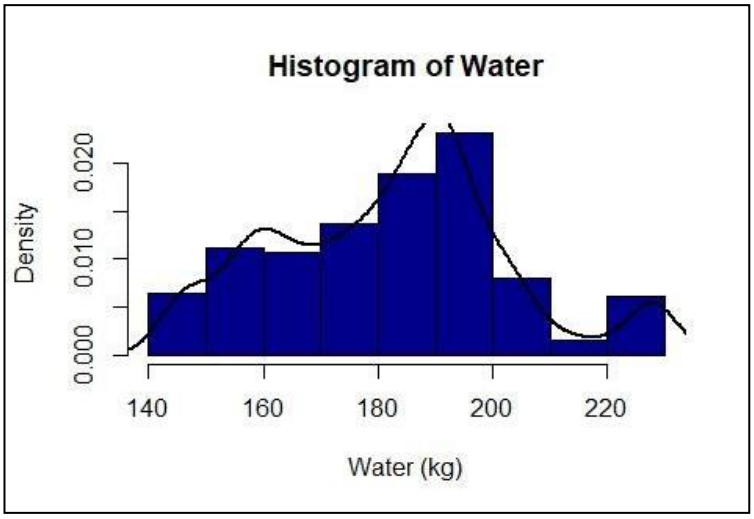


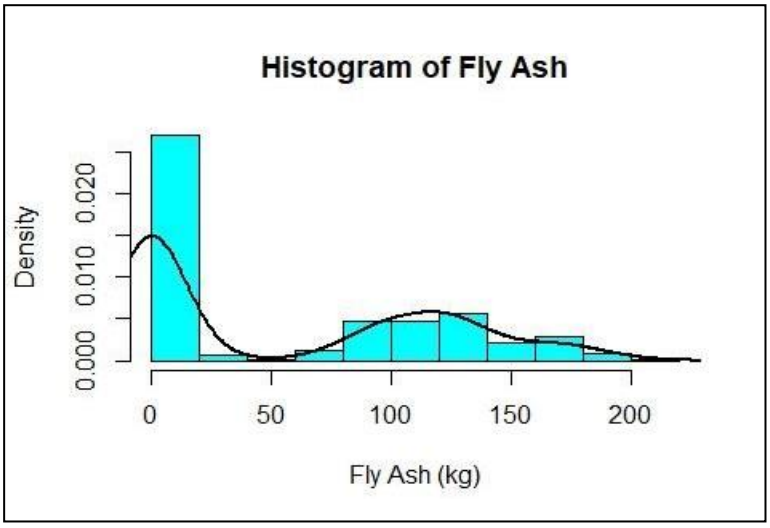
4. Data Visualization

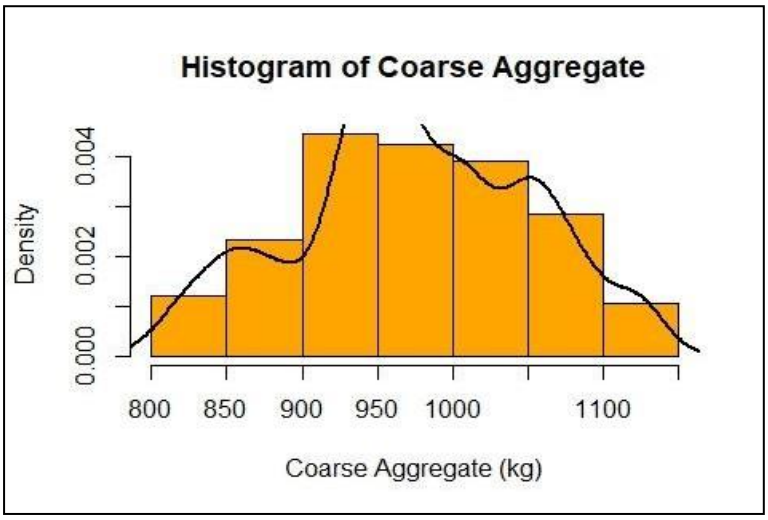
- **Data Visualization** is the graphical representation of information and data. By using visual elements like charts, graphs, maps, and other visual tools, data visualization allows us to see patterns, trends, and outliers in data that may not be obvious in raw numerical formats. It helps to transform complex datasets into a more understandable and actionable form, making it easier to identify insights, communicate findings, and support decision-making processes.

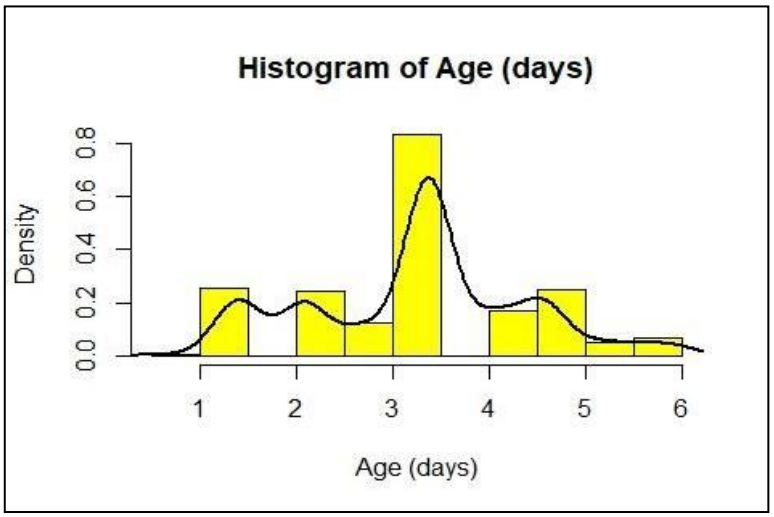
4.1 Histograms

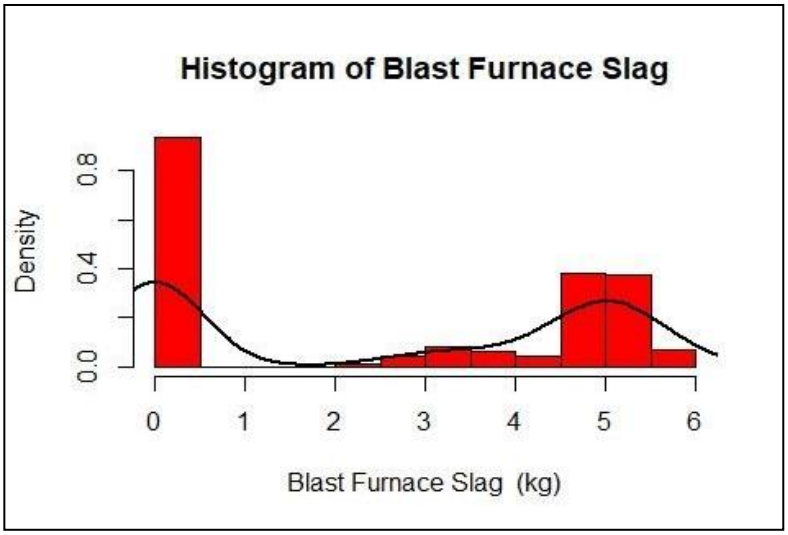
- A histogram is a visual representation of a dataset's distribution, with the x-axis representing the range of values and the y-axis indicating the frequency of data points within each Bin, represented by bars.

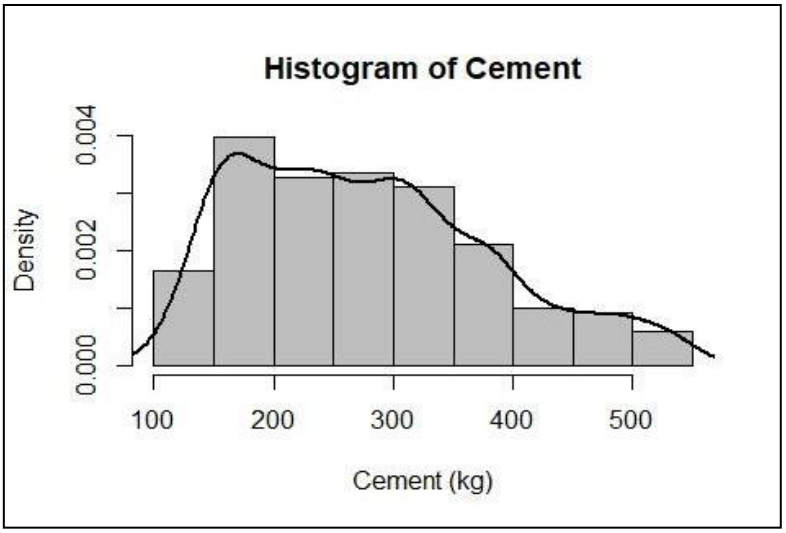


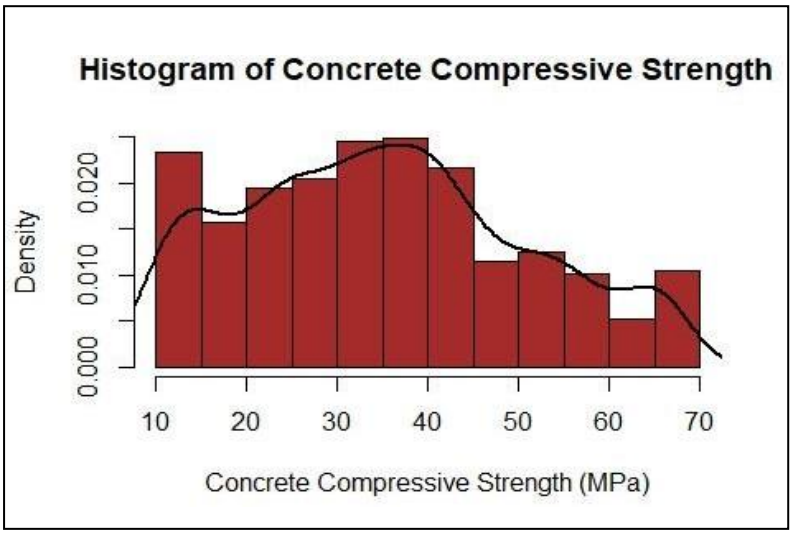


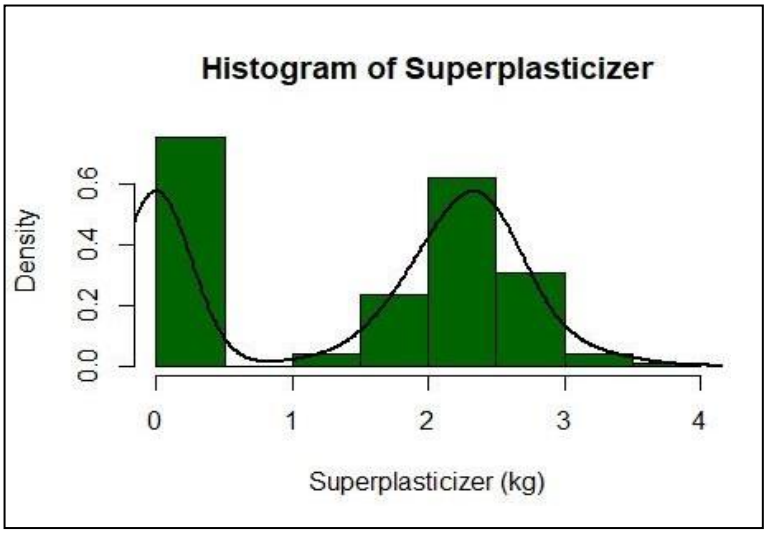


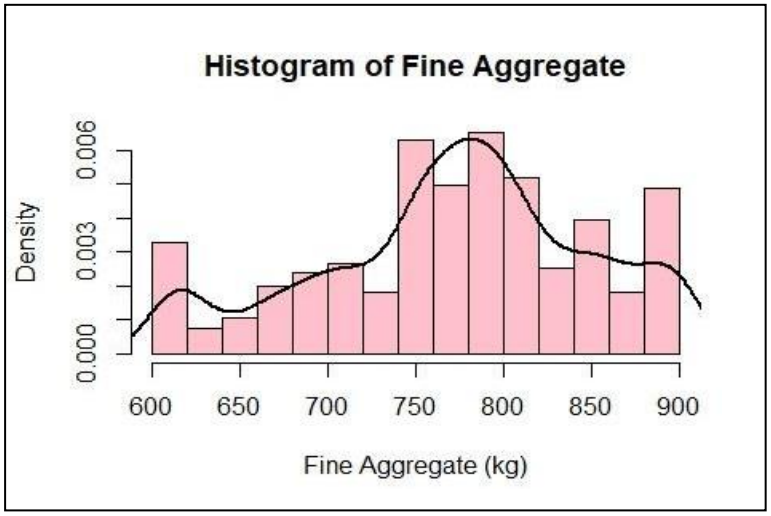



Histogram of Blast Furnace Slag
Density
0.0
0.4
0.8
0
1
2
3
4
5
6
Blast Furnace Slag (kg)


Histogram of Cement
Density
0.000
0.002
0.004
100
200
300
400
500
Cement (kg)

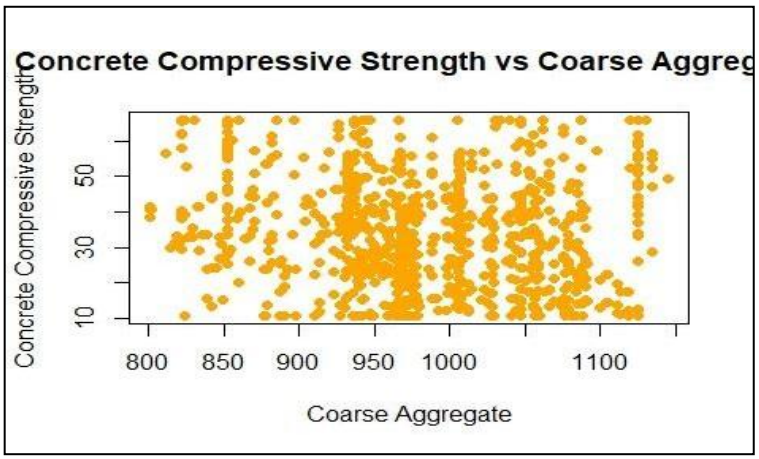

Histogram of Concrete Compressive Strength
Density
0.000
0.010
0.020
10
20
30
40
50
60
70
Concrete Compressive Strength (MPa)

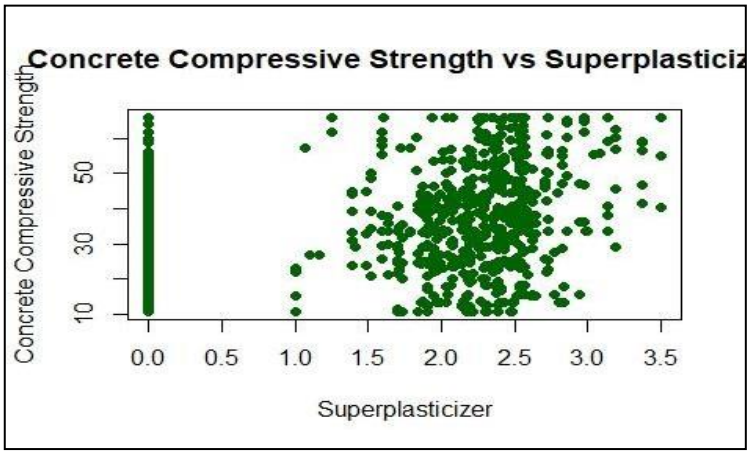

Histogram of Superplasticizer
Density
0.0
0.2
0.4
0.6
0
1
2
3
4
Superplasticizer (kg)

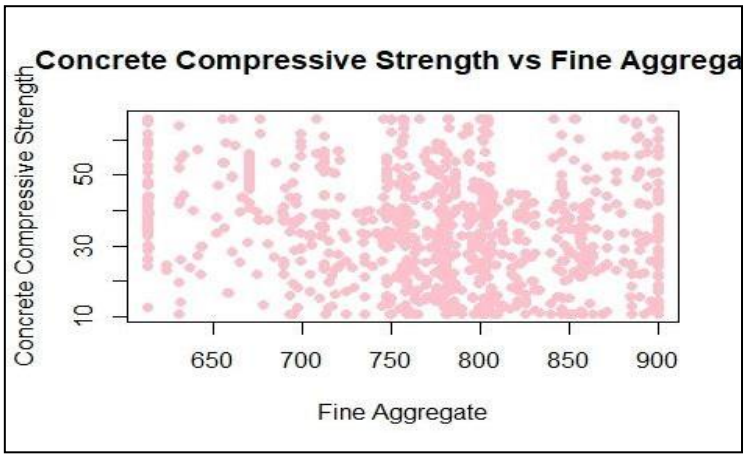

Histogram of Fine Aggregate
Density
0.000
0.003
0.006
600
650
700
750
800
850
900
Fine Aggregate (kg)

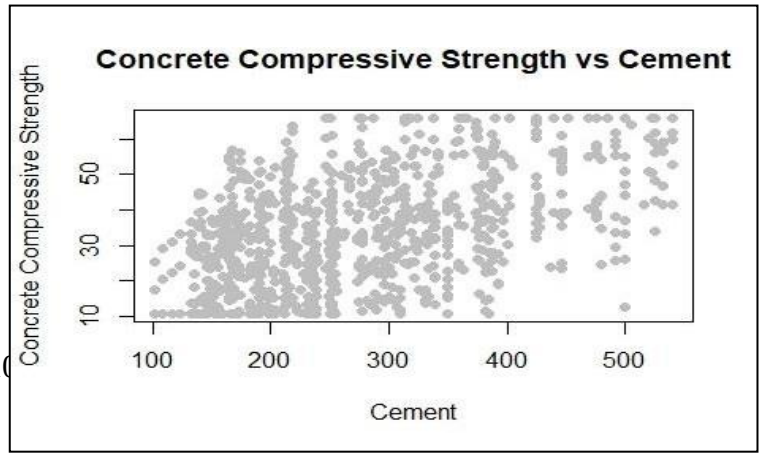
4.2 Scatter Plots

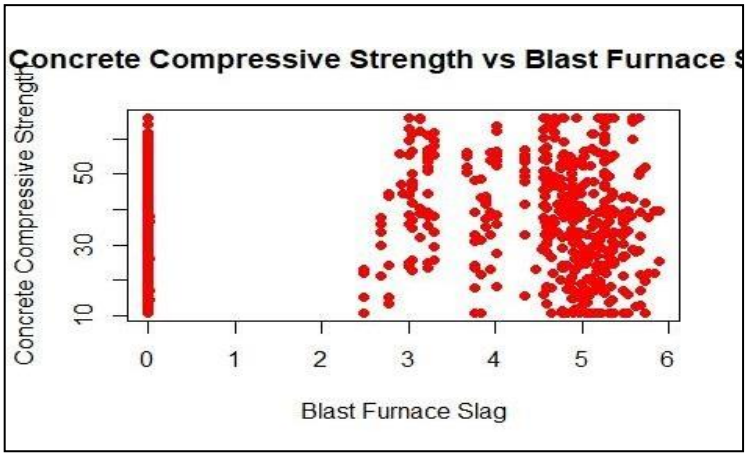
- A scatter plot is a type of data visualization that displays the relationship between two numerical variables by plotting data points on a two-dimensional graph. Each point on the scatter plot represents an observation in the dataset, with one variable plotted along the x-axis and the other along the y-axis. Scatter plots are widely used in data analysis to detect patterns, trends, correlations, and outliers between two variables.

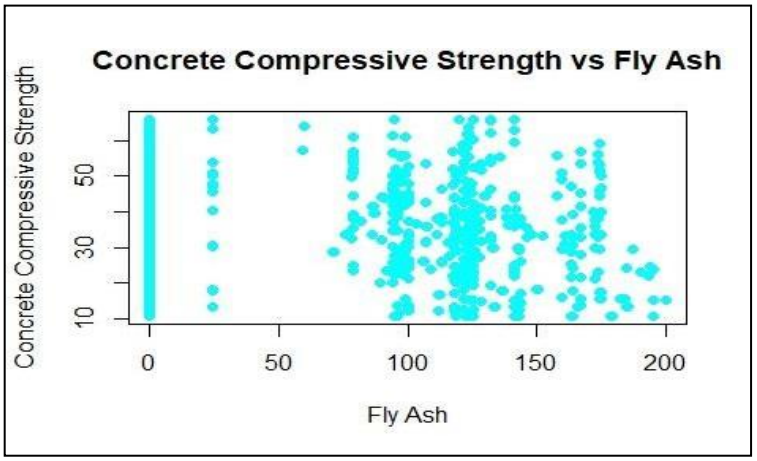


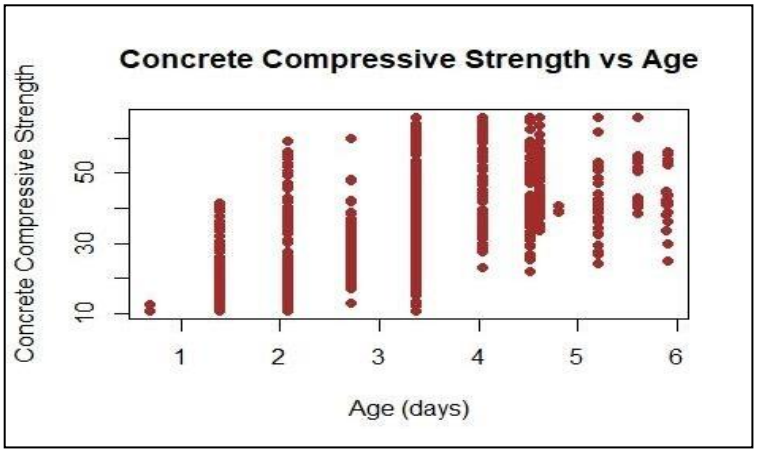


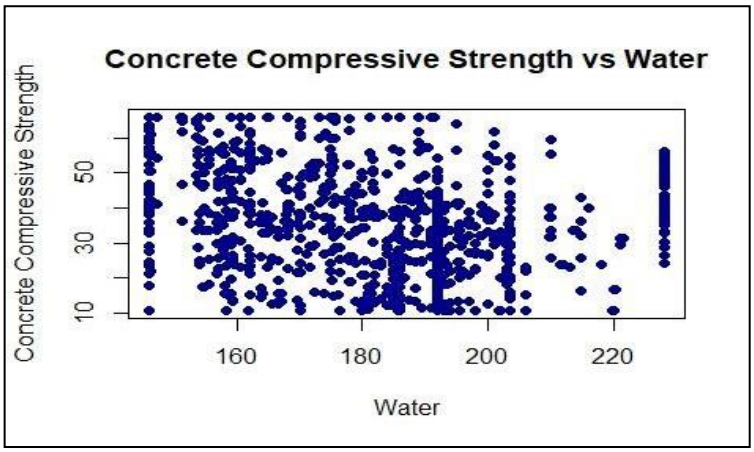





Concrete Compressive Strength vs Blast Furnace S
Concrete Compressive Strength
10
30
50
0
1
2
3
4
5
6
Blast Furnace Slag


Concrete Compressive Strength vs Fly Ash
Concrete Compressive Strength
10
30
50
0
50
100
150
200
Fly Ash


Concrete Compressive Strength vs Age
Concrete Compressive Strength
10
30
50
1
2
3
4
5
6
Age (days)


Concrete Compressive Strength vs Water
Concrete Compressive Strength
10
30
50
160
180
200
220
Water

5. Model Building

5.1 Split the Dataset

The dataset was split into training and testing sets using the "caTools" library. 70% of the data was allocated for training, and 30% was reserved for testing to evaluate the model's performance.

```
#build the model
#split the dataset
library(caTools)
set.seed(123)
split=sample.split(cement_new$con,SplitRatio = 0.7)
training=subset(cement_new,split==T)
testing=subset(cement_new,split==F)
```

5.2 Build the Regression Model

A multiple linear regression model was built on the training data, with compressive strength as the dependent variable and the various concrete components as independent variables.

```
#build the model on training data
con_model=lm(con~cem+blast+ash+water+super+cors+fine+age,data=training)
summary(con_model)
```

5.3 Making Predictions

After building the model, predictions were made using the testing dataset.

```
#make predictions
predict_con=predict(con_model,newdata = testing)
predict_con
```

➢

6. Model Summary

```
> summary(con_model)

Call:
lm(formula = con ~ cem + blast + ash + water + super + cors +
    fine + age, data = training)

Residuals:
     Min       1Q   Median       3Q      Max
-19.0007  -4.3099   0.1088   4.2484  18.4490

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 31.923121  16.593985   1.924   0.0548 .
cem          0.080926   0.004776  16.944  < 2e-16 ***
blast        2.050732   0.204170  10.044  < 2e-16 ***
ash          0.010824   0.007469   1.449   0.1477
water       -0.239150   0.027662  -8.645  < 2e-16 ***
super        1.729594   0.408560   4.233 2.61e-05 ***
cors        -0.001006   0.006103  -0.165   0.8691
fine        -0.014195   0.006847  -2.073   0.0385 *
age          8.671532   0.242474  35.763  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- Cement (0.0809): Significant positive effect on strength.
- Blast Furnace Slag (2.0507): Strong positive effect.
- Water (-0.2391): Significant negative impact.
- Superplasticizer (1.7296): Significant positive effect.
- Age (8.6715): Most influential positive factor.
- Fly Ash, Coarse Aggregate, and Fine Aggregate: Less significant or small effects.

Therefore Multiple Linear Regression Model:

$y = 31.923121+0.080926x_1+2.050732x_2+0.010824x_3-0.239150x_4+1.729594x_5-0.001006x_6-0.014195x_7+8.671532x_8$

7. Model Performance Evaluation

7.1 Coefficient of Determination (R^2)

- R-squared (R^2), is a statistical measure that indicates the proportion of the variance in the dependent variable (in this case, compressive strength) that is explained by the independent variables in the model.

```
Multiple R-squared:  0.8078,    Adjusted R-squared:  0.8056
```

- According to the built model, **80.78%** of the variance in the compressive strength of concrete is explained by the independent variables.

7.2 Analysis of Variances (ANOVA)

- ANOVA evaluates whether the independent variables as a whole explain a significant portion of the variance in the dependent variable.

H0: There is no regression relationship between $x_1, x_2, \ldots\ldots x_8$ and y.

H1: There is a regression relationship between $x_1, x_2, \ldots\ldots x_8$ and y.

OR

H0: $\beta_1 = \beta_2 = \ldots\ldots\ldots = \beta_8 = 0$

H1: $\beta_1 \neq \beta_2 \neq \ldots\ldots\ldots \neq \beta_8 \neq 0$

```
> anova_con=anova(con_model)
> anova_con
Analysis of Variance Table

Response: con
           Df Sum Sq Mean Sq   F value    Pr(>F)    
cem         1  39757   39757  849.3843 < 2.2e-16 ***
blast       1  18106   18106  386.8192 < 2.2e-16 ***
ash         1   8986    8986  191.9832 < 2.2e-16 ***
water       1   6358    6358  135.8310 < 2.2e-16 ***
super       1   2231    2231   47.6702 1.140e-11 ***
cors        1    759     759   16.2246 6.247e-05 ***
fine        1    442     442    9.4451    0.0022 ** 
age         1  59864   59864 1278.9739 < 2.2e-16 ***
Residuals 694  32484      47                       
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

For example, Consider the significance level (α) as 0.05.

According to the above output,

- all independent variables have p-values less than 0.05, we can conclude that each of these variables has a statistically significant effect on compressive strength (concrete strength).

- The model is appropriate, and the independent variables collectively contribute to explaining the variance in the dependent variable.

8. Model Adequacy Checking

The major **assumptions** that we have made to study regression analysis are as follows:

- The relationship between the response *y* and the regressors is linear, at least approximately.
- The error term ε has zero mean.
- The error term ε has constant variance σ^2.
- The errors are uncorrelated.
- The errors are normally distributed.

8.1 Residual Analysis

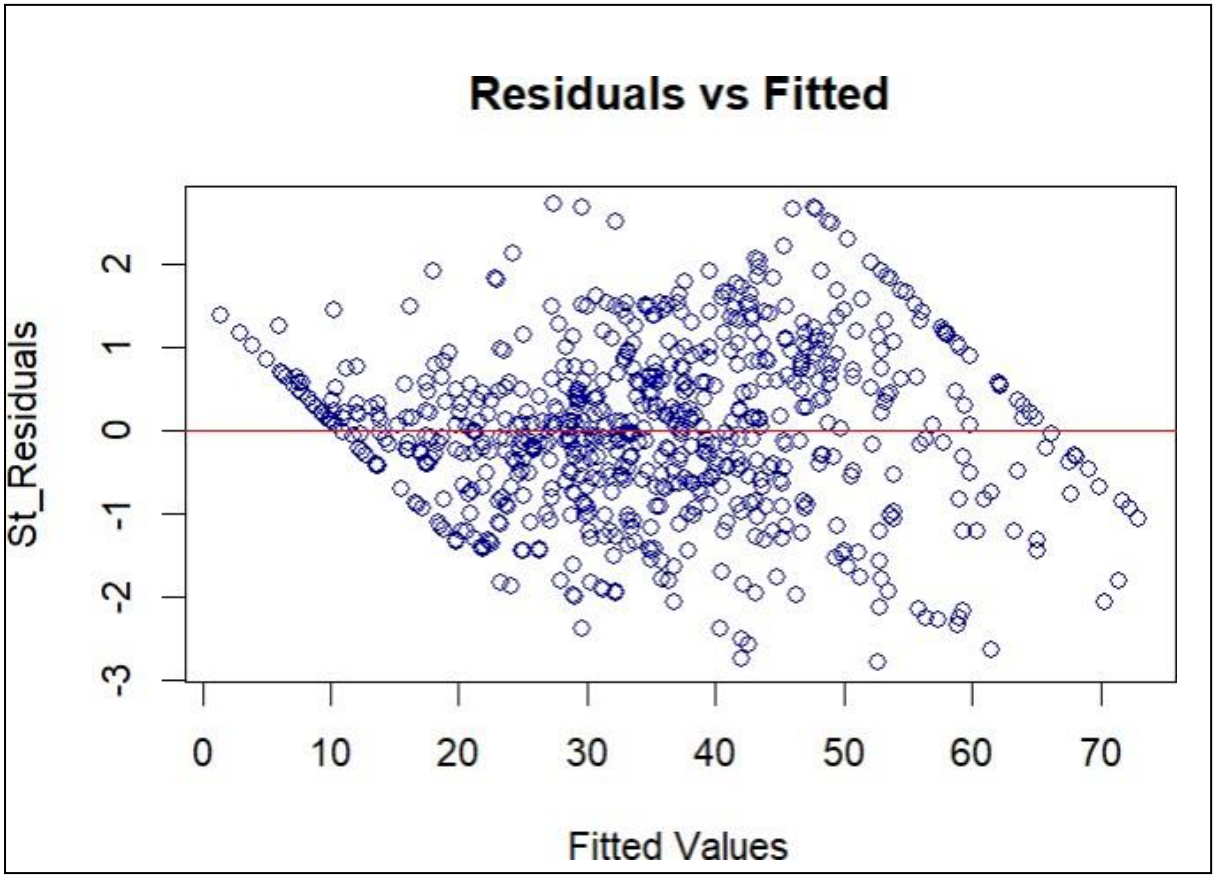
- By examining the differences between observed and predicted values, residual analysis provides information about the adequacy of the model fit.

- Residual analysis helps identify potential issues with the statistical model, such as outliers or violations of assumptions.

- One way to understand residual analysis is by examining the components of a **residual plot.**

- **Residual plots** are graphical representations of the residuals against the predictor variables in a regression analysis.

.



1. Check the linearity of the residuals.

- The residuals show a **curved pattern**, which indicates that there may be a **non-linear relationship** between the independent variables and the dependent variable.

2. Check Heteroscedasticity (Constant variance)

- This indicates **heteroscedasticity**, where the variance of the residuals is not constant across all levels of fitted values. This violates one of the key assumptions of linear regression, which assumes constant variance.

3. Check whether residuals are normally distributed.

3.1 Shapiro Wilk test

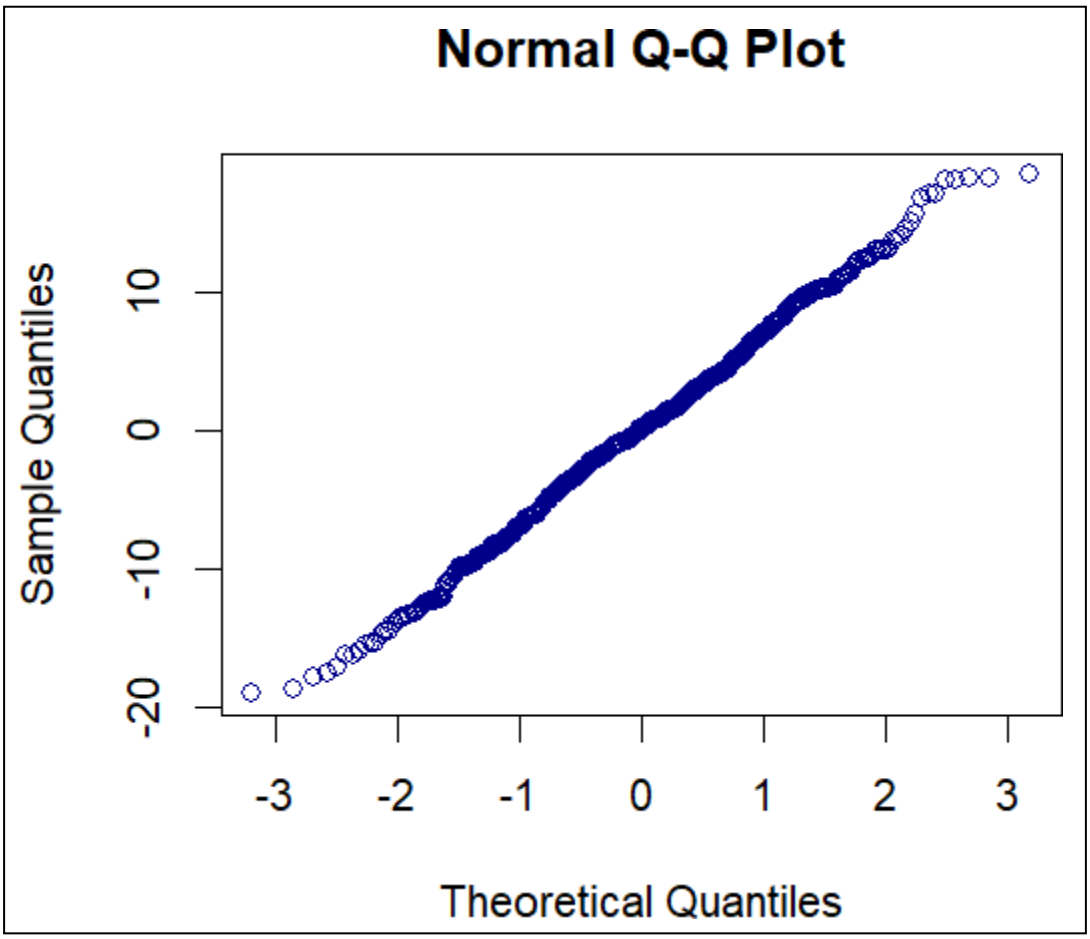
H0: Residuals are normally distributed.
H1: Residuals are not normally distributed.

```
        Shapiro-Wilk normality test

data:  con_model$residuals
W = 0.99713, p-value = 0.2536
```

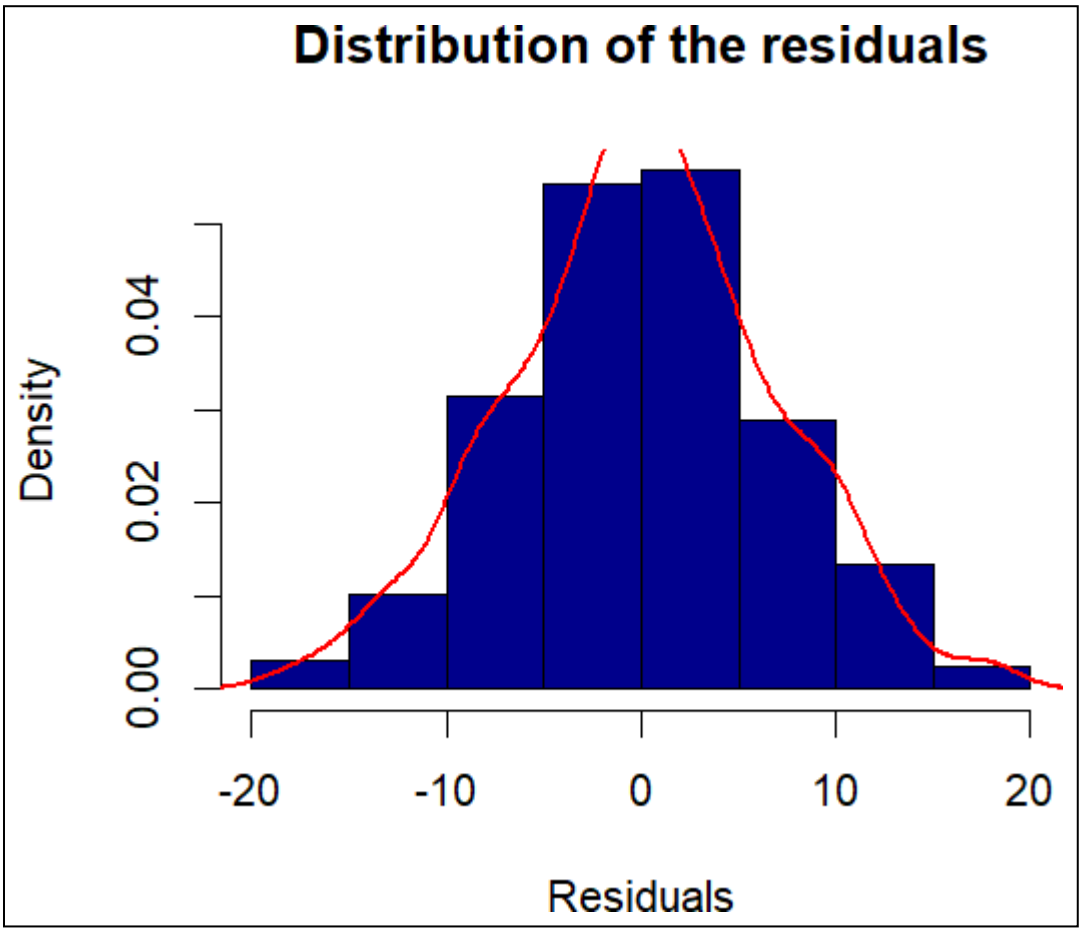
- Since p-value > 0.05, We reject the alternative hypothesis (H1) and we can conclude that residuals are normally distributed.

3.2 Normal Q-Q plot



- The plot shows that residuals closely follow a straight diagonal line which indicates that residuals are normally distributed.

3.3 Visualizing using a Histogram



4. Correlation among residuals.

- We can use **Durbin Watson Test** to check whether independence among residuals

H0: There is no correlation in the residuals.
H1: There is a correlation in the residuals.

```
        Durbin-Watson test

data:  con_model
DW = 1.4046, p-value = 3.084e-16
alternative hypothesis: true autocorrelation is greater than 0
```

- Since p-value < 0.05, We do reject the null hypothesis (H0) and we can conclude that there is a correlation in the residuals.

8.2 Multicollinearity

- We expect to find dependencies between the response variable and regressors.

- There are also dependencies among the regressor variables.

- In situations where these dependencies are strong, we say that **multicollinearity** exists.

    1. **Variance Inflation Factor (VIF)** is a metric used to detect **multicollinearity** in regression analysis. Multicollinearity occurs when two or more independent variables (predictors) in a model are highly correlated, meaning they provide redundant information about the response variable. This can affect the stability and interpretation of the model's coefficients.

        VIF = 1: No correlation between the independent variable and the other Variables.

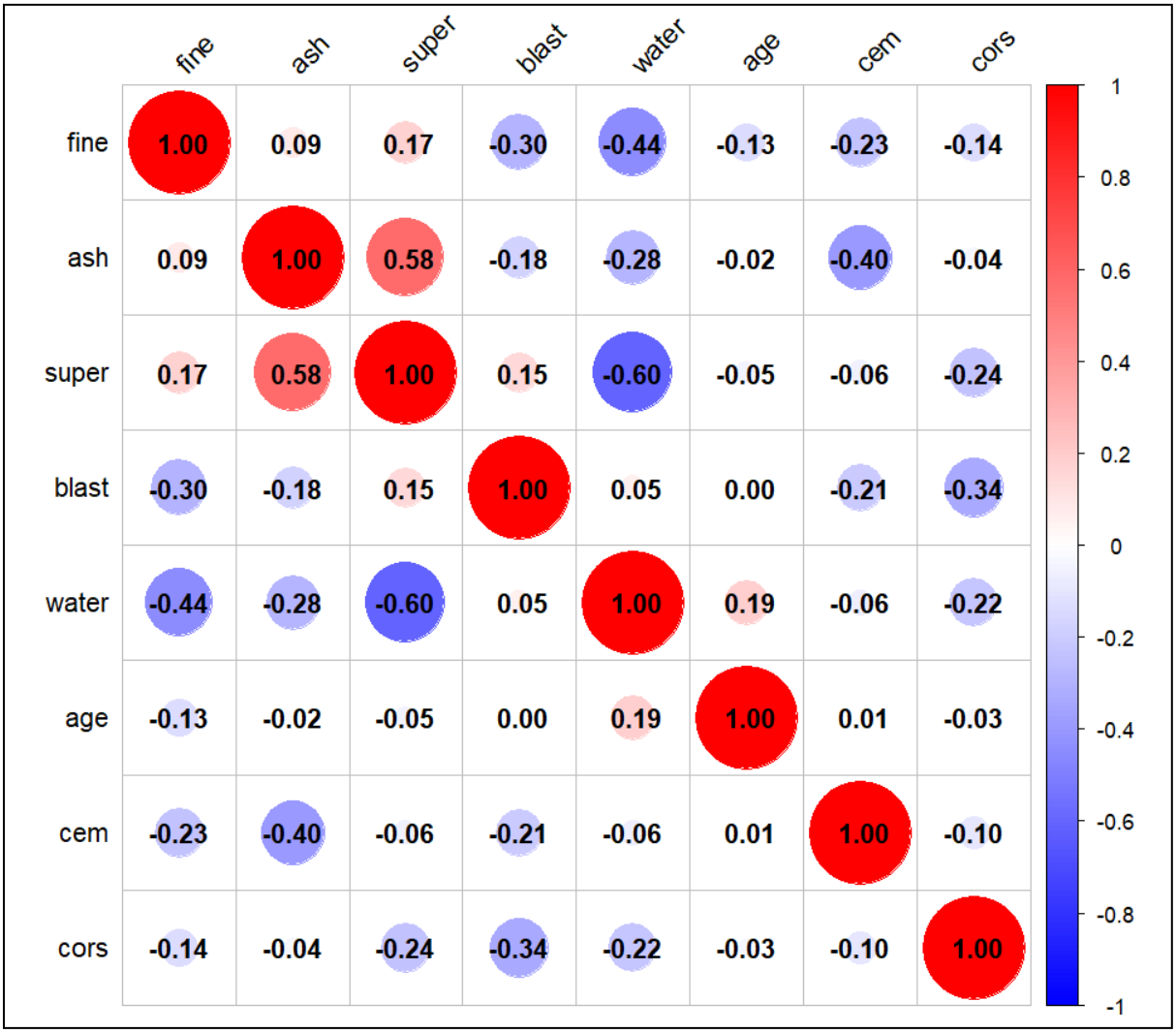
        1 < VIF < 5: Moderate correlation; generally considered acceptable.

        VIF > 5 or 10: High multicollinearity.

```
> vif(con_model)
     cem    blast      ash    water    super     cors     fine      age
3.601756 3.572321 3.517477 4.867627 3.298421 3.390313 4.115722 1.050403
>
```

- Since all the **VIF values fall within 1 to 5** it indicates that there is a **moderate correlation** among independent variables. It can be generally acceptable.

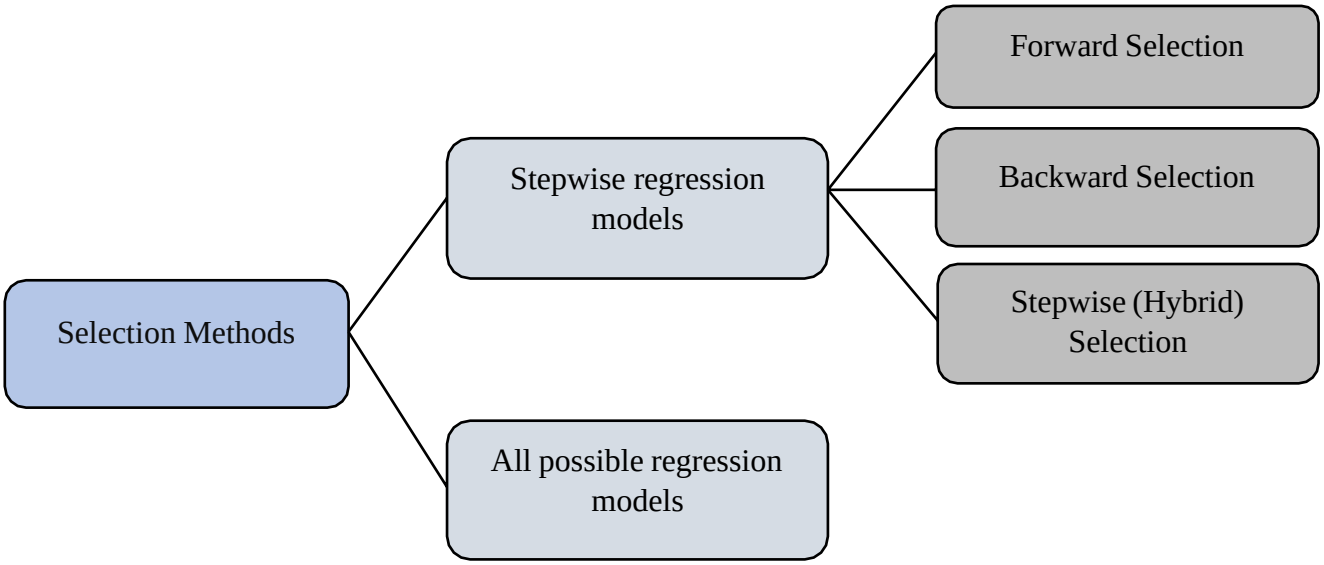
2. **Correlation Matrix:** Correlation between independent variables using a correlation matrix.



9. Variable Selection Methods

- Variable selection refers to choosing the most relevant variables in a regression model. It helps improve model performance and avoid overfitting.

- There are two variable selection methods.



9.1 Stepwise Selection

- Stepwise regression is a method of fitting regression models that involves the iterative selection of independent variables to use in a model. It can be achieved through forward selection, backward elimination, or a combination of both methods.

- The forward selection approach starts with no variables and adds each new variable incrementally, testing for statistical significance, while the backward elimination method begins with a full model and then removes the least statistically significant variables one at a time.

- In here we only consider the **stepwise backward selection**.

Stepwise Summary

Step	Variable	AIC	SBC	SBIC	R2	Adj. R2
0	Full Model	4709.725	4755.279	2714.931	0.80777	0.80556
1	cors	4707.753	4748.751	2712.932	0.80777	0.80583

- Step 0 represents the full model, including all predictor variables. R^2 and adjusted R^2 show that the model explains around 80% of the variance in the response variable.

- Step 1 "cors" variable is selected, which seems to slightly improve the model, as the AIC and SBC decreased a bit compared to the full model, suggesting a better fit.

ANOVA

	Sum of Squares	DF	Mean Square	F	Sig.
Regression	136502.435	7	19500.348	417.199	0.0000
Residual	32485.087	695	46.741		
Total	168987.522	702			

- The p-value is extremely small (less than 0.05), indicating that the model is statistically significant overall.

Parameter Estimates

model	Beta	Std. Error	Std. Beta	t	Sig	lower	upper
(Intercept)	29.420	6.685		4.401	0.000	16.295	42.545
cem	0.081	0.004	0.539	22.570	0.000	0.074	0.089
blast	2.074	0.150	0.319	13.855	0.000	1.780	2.367
ash	0.011	0.007	0.047	1.672	0.095	-0.002	0.025
water	-0.236	0.018	-0.313	-12.941	0.000	-0.271	-0.200
super	1.752	0.385	0.130	4.545	0.000	0.995	2.509
fine	-0.013	0.005	-0.066	-2.872	0.004	-0.023	-0.004
age	8.671	0.242	0.610	35.791	0.000	8.195	9.146

Interpretation

- The model appears to be well-fitted, with an R^2 of 0.808, a high F-Statistic, and significant predictors.
- The variables like cem, blast, water, super, and age are significant and have a substantial impact on the response variable. They should be retained in the model.
- The ash variable is not statistically significant.

9.2 Evaluating Subset Regression Model

1. **AIC (Akaike Information Criterion)**

- AIC is a measure used to compare the goodness of fit of different statistical models, while also penalizing the complexity of the model.
- The goal of AIC is to balance the trade-off between the model's fit and its complexity, ensuring that we do not overfit the data by using too many predictors.

2. **BIC (Bayesian Information Criterion)**

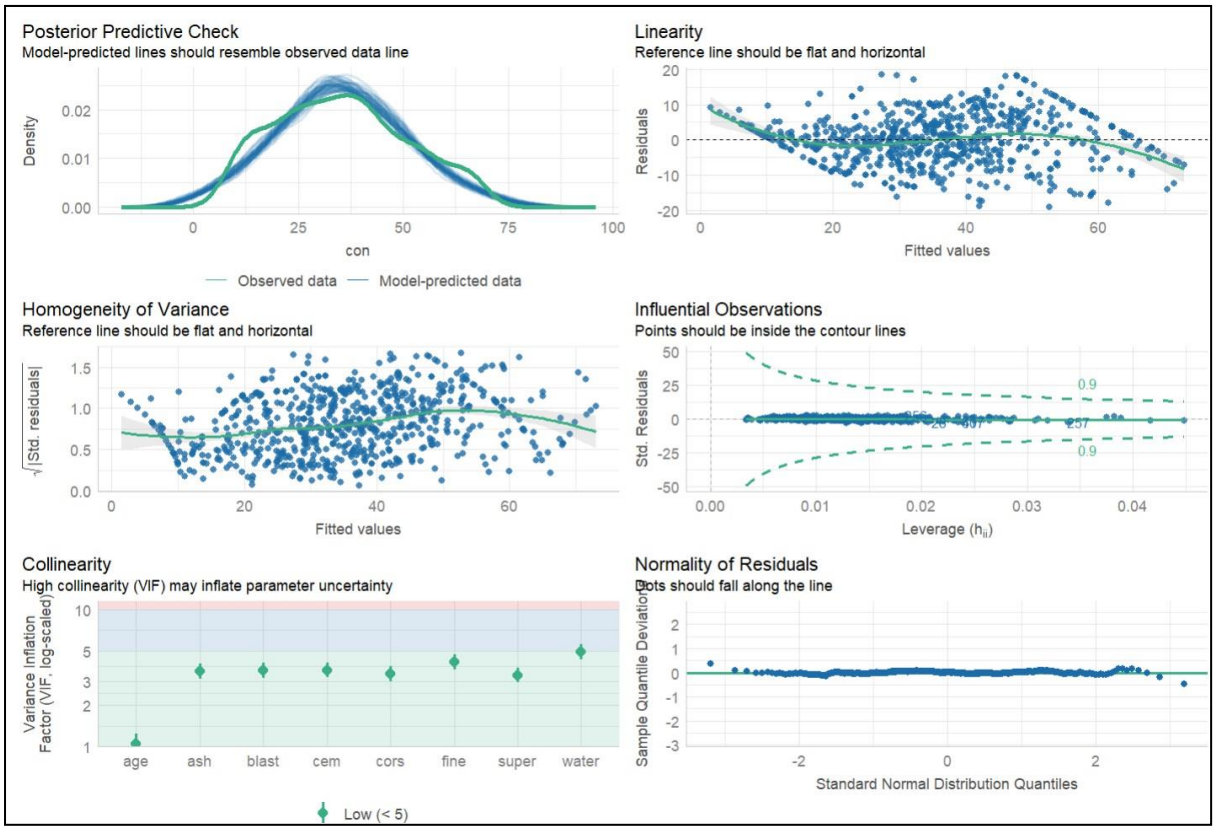
- BIC also known as the Schwarz Bayesian Criterion (SBC), is another criterion for model selection among a finite set of models.
- It aims to balance model fit and complexity, but it applies a different penalty for the number of parameters.

```
      model      AIC      BIC
1 con_model 4709.725 4755.279
2  back_mod 4708.574 4745.017
```

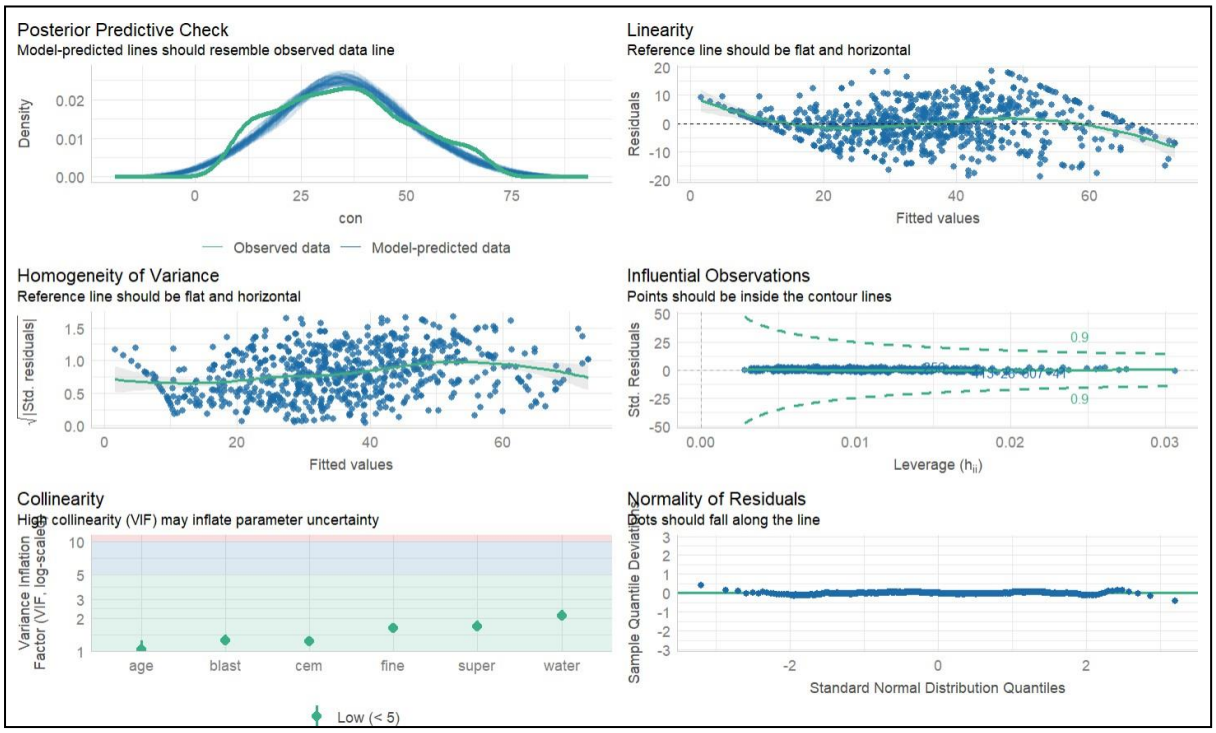
- According to AIC and BIC suggest that **back_mod** is fitted for further analysis.

10. Visualizing the overall performance of the Regression models

10.1 Concrete model (with all the predictor variables)



10.2 Stepwise Backward Selection



11. Conclusion

- In conclusion, this study effectively developed a multiple linear regression model to predict the compressive strength of concrete based on several key factors, including cement, water, blast furnace slag, superplasticizer, and the curing age of the concrete. The model showed **strong performance**, explaining **approximately 80%** of the variation in compressive strength, which indicates a good fit for making predictions.

- Significant factors, such as **cement and blast furnace slag**, were found to have a **positive impact on** the strength, while **water** had a **negative effect**. The **age of the concrete** was the **most influential factor**, demonstrating that as concrete cures, its strength increases.

- However, the analysis revealed a few issues. Residual analysis showed that the model might not fully capture non-linear relationships between variables, and **there was evidence of heteroscedasticity**, meaning that the variance of errors changes at different levels of the predicted values. These factors suggest that the model could be improved by incorporating techniques to better handle non-linear effects or by exploring alternative models.

- Despite these limitations, the current model is a valuable tool for predicting concrete compressive strength, providing useful insights for engineers and builders to make informed decisions. By refining the model further, even more accurate and reliable predictions could be achieved, enhancing construction safety and material performance evaluation.

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